

Density - Based Clustering :

↳ Unsupervised machine learning

↳ Based on distance between nearest points.

↳ It identifies distinctive groups / clusters in data space is a contiguous region of high points density. clusters

↳ Noise or outliers separated from other continuous dense points with low density. noise

↳ Thus density based clustering is based on the idea that clusters are regions of high density separated by region of low density.

↳ can find arbitrary shape clusters.

↳ while partitioning and hierarchical clustering are designed to find spherical - shaped clusters.

↳

working of Density Based Clustering

↳ The algorithm works by first identifying core points, which are data points that have a ~~min~~ minimum no. of neighbors within specified distance. These core points form the center of a cluster.

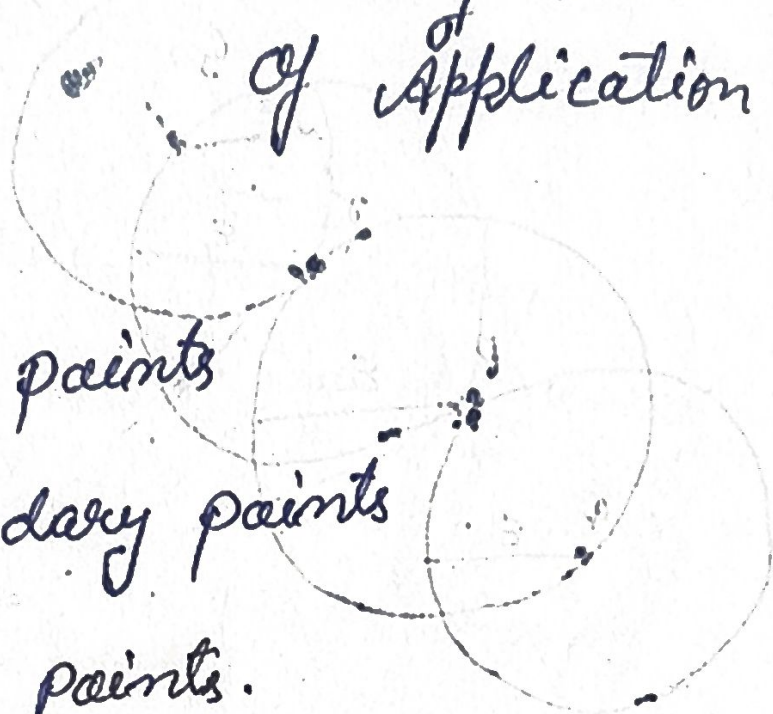
↳ Next the algorithm identifies border data points, which are data points that are not core points but have at least one core data points as a neighbor.

↳ Finally, the algorithm identifies "noise" points which are data points that are not core data points or border data points.

Density Based clustering:

DBSCAN (Density Based Spatial Clustering of Application with noises).

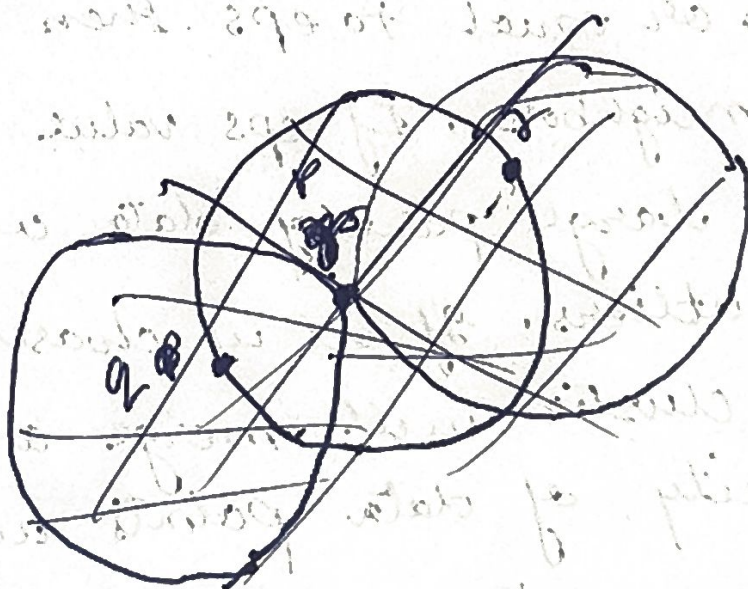
- ① Core points
- ② Boundary points
- ③ Noise points.



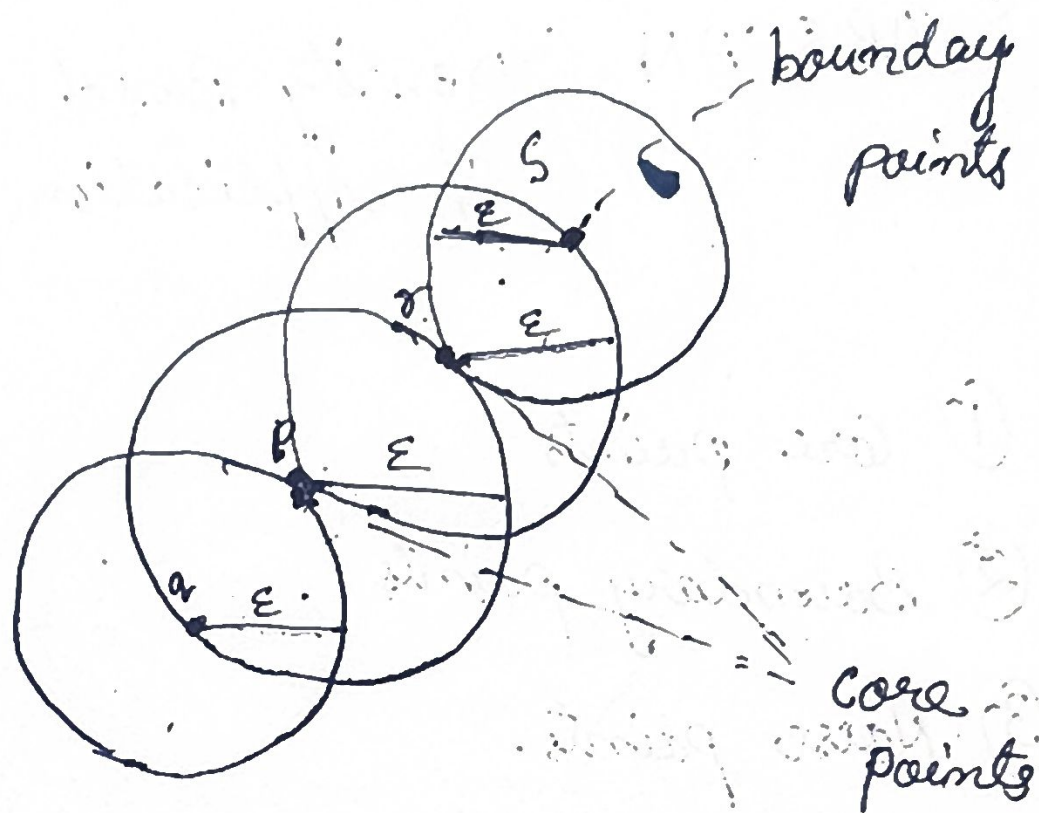
Working principal

① Eps

② MinPts = 3



Months : 3



Direct Density reachable.

Parameters required for DBSCAN algorithm.

1. ϵ : It defines the neighborhood around a data point. i.e. if the distance between two points is lower or equal to ϵ then they are considered neighbors. If ϵ value is chosen too small then large part of data will be considered as outliers. If it is chosen very small large then cluster will merge and the majority of data points will be the same clusters. One way to find the ϵ value is based on k -distance graph.

2. MinPts: Minimum number of neighbors (data points) within ϵ s radius. The larger the dataset, the larger value of MinPts must be chosen. As a general rule, the minimum MinPts can be derived from the number of dimensions D in the dataset as $\text{MinPts} \geq D+1$. The minimum value of MinPts must be chosen at least 3.

Three Data Points:

Core Point: A point is a core point if it has more than MinPts within ϵ s.

Border Point: A point which has fewer than MinPts within ϵ s but it is in the neighborhood of a core point.

Noise / Outliers: A point which is not core point or border point.

Two Concepts:

1. Reachability: Reachability in terms of density establishes a point to be reachable from another if it lies within a particular distance (ϵ s) from it.

2. Connectivity : Connectivity on the other hand, involves a transitivity based chaining approach to determine whether points are located in a particular cluster. For example, p and q points could be connected if $p \rightarrow r \rightarrow s \rightarrow t \rightarrow q$ where $a \rightarrow b$ means b is in the neighborhood of a .

Steps Used in DBSCAN algorithm.

1. Find all the neighbor points within ϵ and identify the core points or visited with more than $MinPts$ neighbors.
2. For each core point if it is not already assigned to a cluster, create a new cluster.
3. Find recursively all its density-connected if there exists a point c which has sufficient number of points in its neighbor and both points a and b are within the ϵ distance. This is chaining process. So if b is neighbor of c , c is neighbor of d and d is neighbor of a , which in turn is neighbor of a implying that b is neighbor of a .
4. Iterate through the remaining unvisited points in the dataset. Those points do not belong to any cluster are noise.